Report: Analysis of Building Vibration Data

Date: 2025-10-07

Executive Summary

This report details the analysis of accelerometer data collected from a building to determine its fundamental natural frequency. By applying signal processing techniques to the provided time-series data, the analysis identified a clear primary frequency at which the structure tends to oscillate.

The estimated natural frequency of the building is **2.160 Hz**.

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1. Data

- **Source File:** `Copy of QuakeData_ShortNoSnow35% - Copy of QuakeData_ShortNoSnow35%.csv`
- **Data Type:** Time-series data from an accelerometer.
- **Structure:** The data consists of three columns: `Time (s)`, `A-output`, and `B-output`. For this analysis, the `A-output` column was used.
- **Sampling Rate:** The time column data increases in increments of 0.005 seconds, which yields a sampling frequency (Fs) of **200 Hz** (1 / 0.005).

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2. Methodology

The analysis was performed using a custom Octave script (`analyze_quake_data.m`). The process consisted of the following steps:

2.1. Data Loading and Visualization

The script first loaded the accelerometer data from the CSV file. The raw vibration data (`A-output`) was plotted against time to provide a visual representation of the building's oscillations.

2.2. Frequency-Domain Transformation

To identify the component frequencies within the signal, a **Fast Fourier Transform (FFT)** was performed. This mathematical process transforms the time-domain signal into a frequency-domain representation, showing the amplitude of each frequency present in the signal.

Prior to the FFT, the signal was detrended by removing its mean value (DC offset). This centers the signal around zero and prevents a large, erroneous peak at 0 Hz in the frequency spectrum.

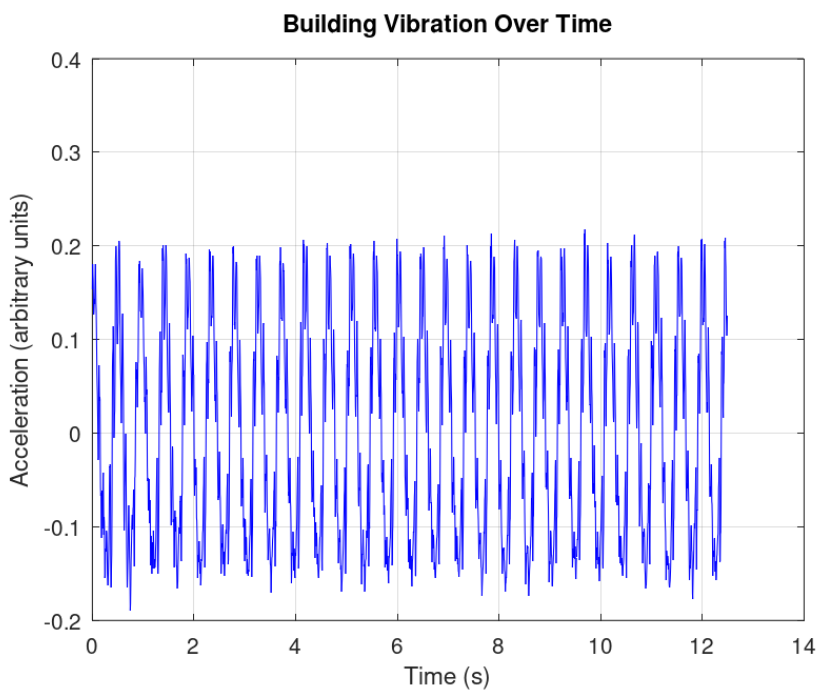

Figure 1: Building Vibration Over Time

2.3. Natural Frequency Identification

The result of the FFT is a power spectrum. The natural frequency of the building manifests as a prominent peak in this spectrum, indicating the frequency at which the building vibrates most strongly. The script identified the frequency corresponding to the highest peak in the spectrum (ignoring the near-zero DC component).

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3. Results

The analysis successfully identified the building's natural frequency.

- **Estimated Natural Frequency: 2.160 Hz**

This result is supported by the two plots generated by the script:

1. **Accelerometer Time-Series Data:** This plot shows the complex oscillations of the building as recorded by the sensor.
2. **Frequency Spectrum Analysis:** This plot clearly shows a single, dominant peak in the frequency domain. A vertical line marks this peak, which is located at 2.160 Hz.

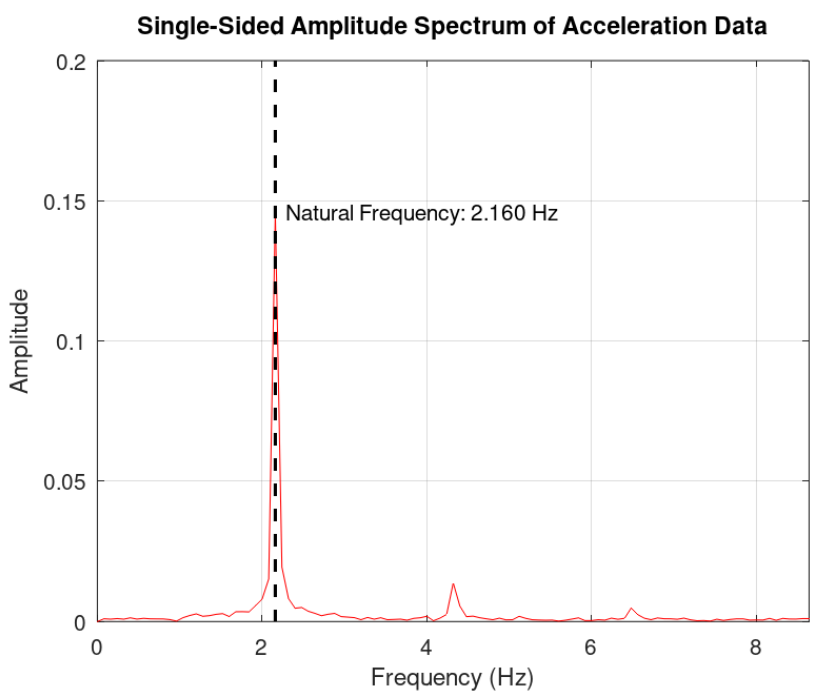

Figure 2: Frequency Spectrum Analysis

4. Conclusion

The application of the Fast Fourier Transform proved to be an effective method for determining the natural frequency of the building from the provided accelerometer data. The result of 2.160 Hz represents a key structural characteristic of the building.